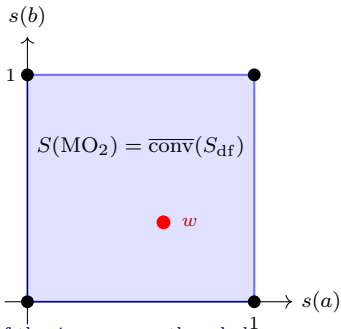


(1) MO₂

corners fill the state space



hull of the 4 corners = the whole square;

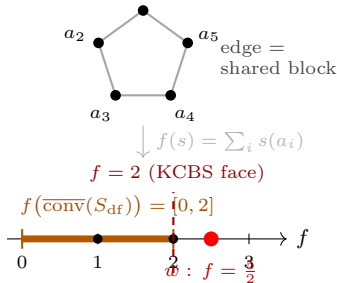
every state is a mixture of corners

⇒ **no contextual state**

(non-distributive, yet non-contextual:
contextuality needs a loop, not just
incompatibility — contrast panel 2)

(2) Pentagon

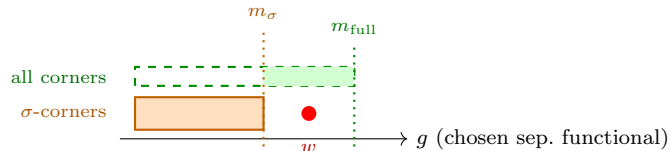
the lattice, and its hull projected



every dispersion-free corner has $f = \sum s(a_i) \leq 2$
(odd loop: no two adjacent atoms both 1), so every
mixture does; but $w \equiv \frac{1}{2}$ has $f = \frac{5}{2} \Rightarrow$ **contextual**

(3) σ -essential

the leak between two hulls



the σ -additive corners reach only $g \leq m_\sigma$; the
extra “fake” corners push the full hull to $g \leq m_{\text{full}}$.
 w has $m_\sigma < g(w) \leq m_{\text{full}}$: **outside** $\text{conv}(S_{\text{df}}^\sigma)$,
inside $\text{conv}(S_{\text{df}})$ — and no finite face sees it

The spaces. The lattice L (e.g. the pentagon: 5 atoms + loop relations) is an order-theoretic object, not a vector space; it indexes the others. A *state* assigns a number to each element, so states live in $\mathbb{R}^L$ — for the pentagon, pinned by the 5 atom-values, a polytope $S(L) \subseteq [0, 1]^5 \subseteq \mathbb{R}^5$. The dispersion-free states S_{df} are its $\{0, 1\}$ -valued *vertices*; $\text{conv}(S_{\text{df}})$ is a sub-polytope, *also in* $\mathbb{R}^5$ — high-dimensional and undrawable. **The projection.** Panels (2)–(3) draw not that polytope but its image under a linear functional $g : \mathbb{R}^L \rightarrow \mathbb{R}$ (a weighted sum of atom-values; in (2), $g = f = \sum_i s(a_i)$). Since g is linear, $g(\text{conv}(S_{\text{df}}))$ is an *interval*, and $g(w)$ outside it forces w outside the hull — a one-way certificate. **Caveat.** g is a *chosen* separating functional for this w , not a canonical projection: by Hahn–Banach, w is contextual *iff some* such g sends it past the hull’s image; one g detects the states violating *that* inequality, not all contextual states.